Sri Lanka Institute of Information Technology

IT3021 - Data warehousing and

Business Intelligence

Assignment 02

IT23672550

Wijesuriya W.M.S.N

Step 1: Data source for the Assignment 2

1.1 Description of Data Source

The data source used for this assignment is the data warehouse developed in Assignment 1, which is based on the NYC 311 Service Requests dataset. The data warehouse is designed using a star schema model to support efficient analytical processing.

The central table in the schema is the Fact_Service_Request table, which stores quantitative data related to service requests. This fact table is connected to several dimension tables that provide descriptive attributes.

Fact Table

Fact_Service_Request

- Stores measurable data (facts)
- Measures include:
    - Request Count
    - Fact Service Request Count
    - Transaction Processing Time Hours

These measures are used for aggregation and analysis in the cube.

Dimension Tables

Dim Date

- Contains time-related attributes such as:
    - Year Number
    - Quarter Number
    - Month Name
    - Full Date
- Used for time-based analysis

Dim Agency

- Contains agency-related identifiers (Agency Key)
- Used to analyze requests by responsible organizations

Dim Complaint

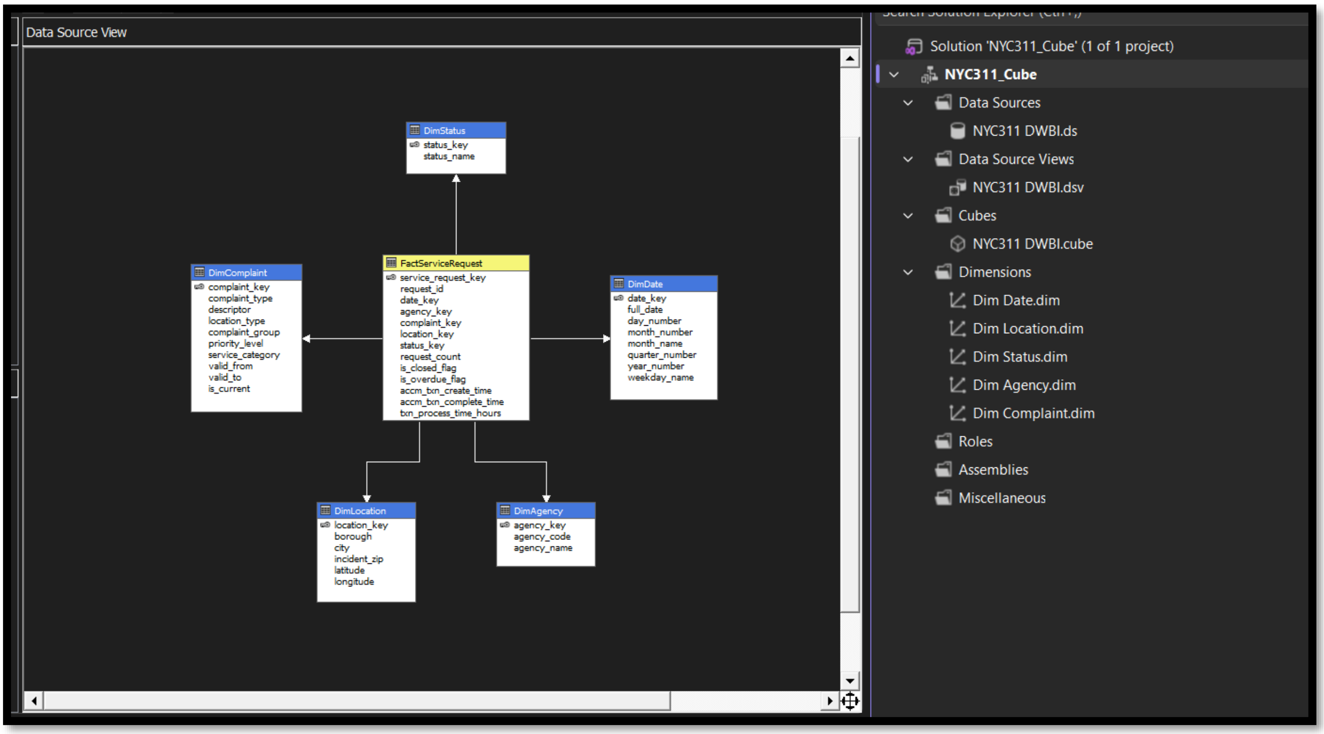
- Contains complaint identifiers
- Used to categorize types of service requests

Dim Location

- Stores location-based data
- Enables geographical analysis

Dim Status

- Represents the status of service requests

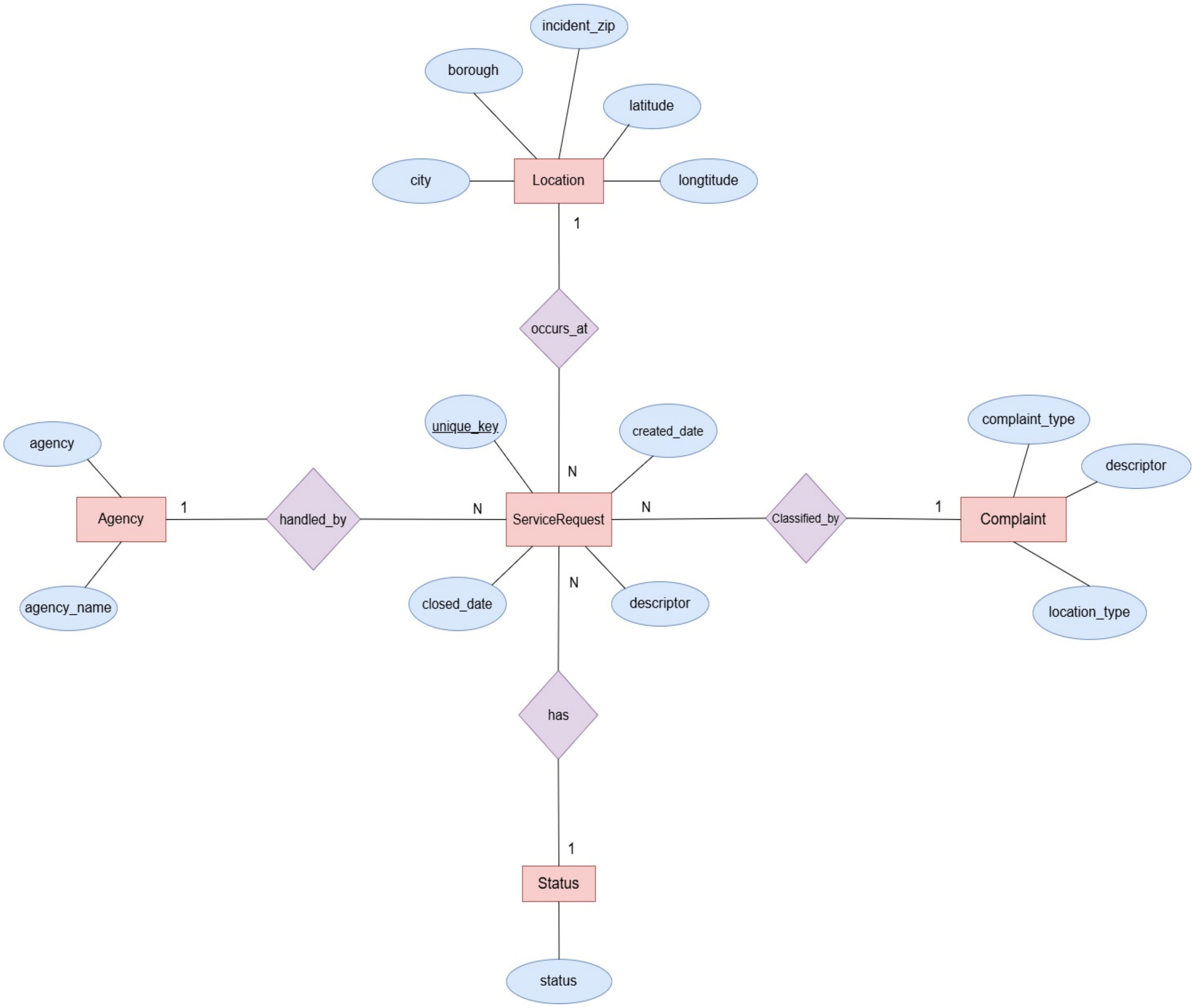


1.2 Design Justification

The star schema structure separates facts and dimensions, allowing:

- Faster query performance
- Easier aggregation
- Better support for OLAP operation

ER Diagram



Conceptual ER diagram for the NYC 311 service request dataset

Step 2: SSAS Cube Implementation

2.1 Cube Creation

An SSAS Multidimensional Cube was created using the data warehouse as the data source. The cube enables multidimensional analysis by organizing data into measures and dimensions.

Steps Followed:

1. Created a new Analysis Services Project in Visual Studio
2. Defined Data Source connecting to the data warehouse
3. Created Data Source View (DSV) including fact and dimension tables
4. Created a new Cube using Cube Wizard
5. Selected:
   - Fact table: FactServiceRequest
   - Measures: Request Count, Processing Time
6. Added dimensions:
   - Dim Date
   - Dim Agency
   - Dim Complaint
   - Dim Location
   - Dim Status
7. Processed and deployed the cube

2.2 Measures and Dimensions

- Measures were selected from the fact table to support aggregation (e.g., Request Count)
- Dimensions were created from dimension tables to provide analytical context

2.3 Hierarchy Implementation

A time hierarchy was implemented in the Dim Date dimension:

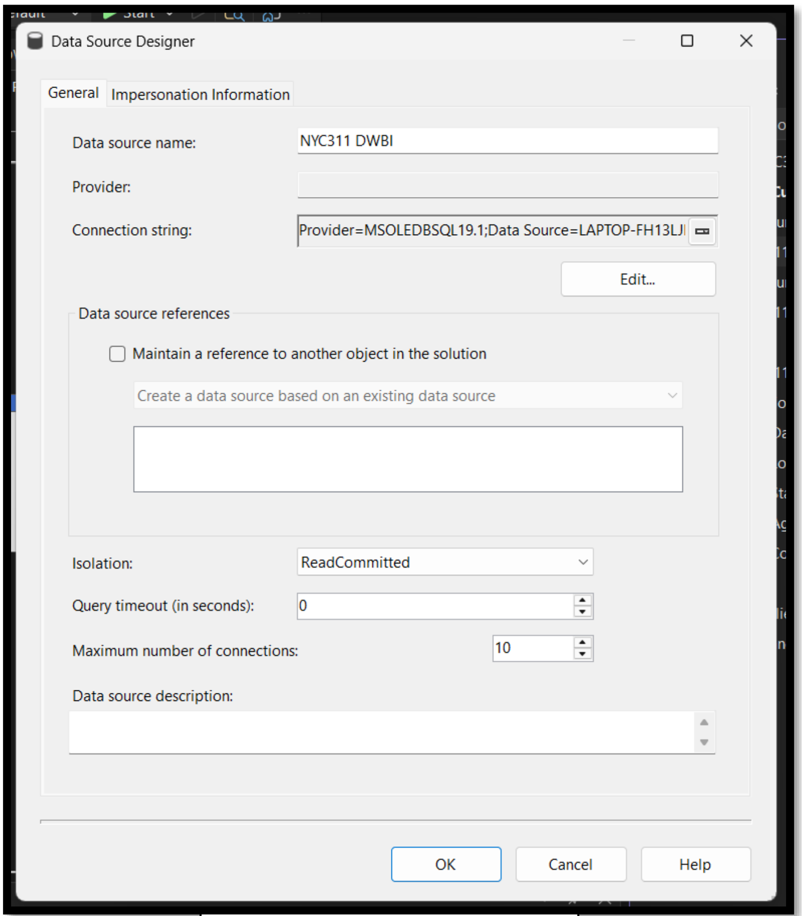
Year -> Quarter -> Month -> Date

This hierarchy enables:

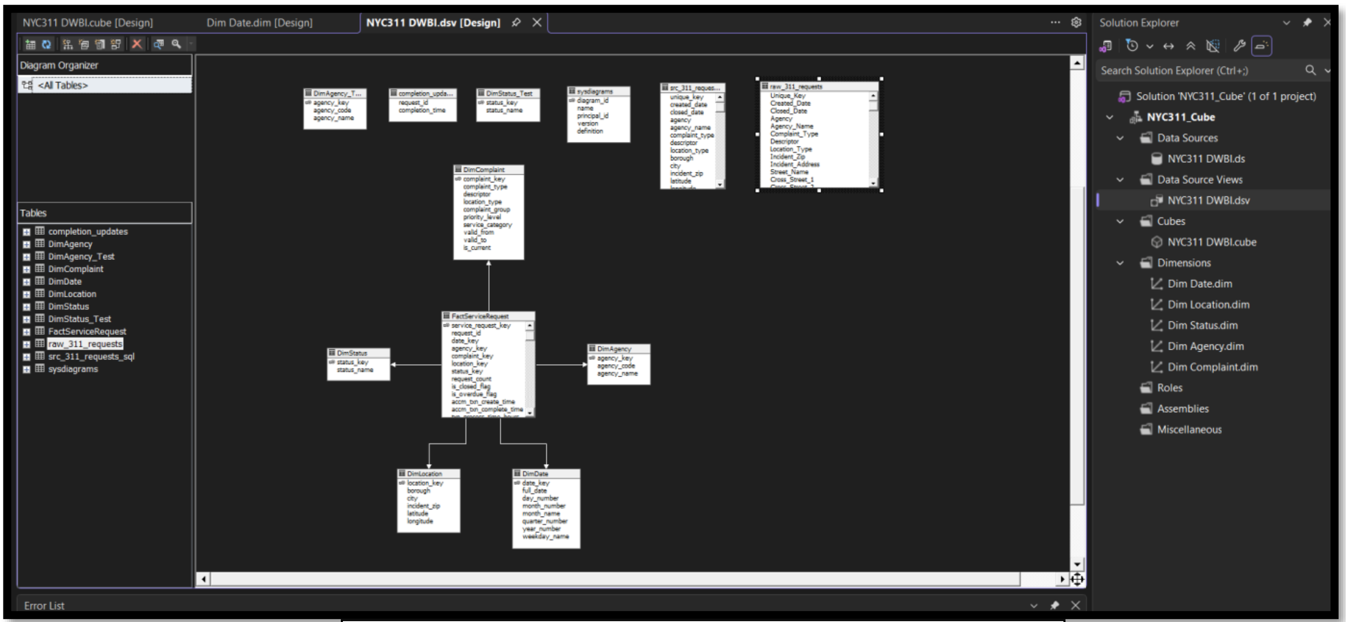
- Drill-down and roll-up operations
- Multi-level time analysis
- Improved user interaction and performance

2.4 Cube Design Summary

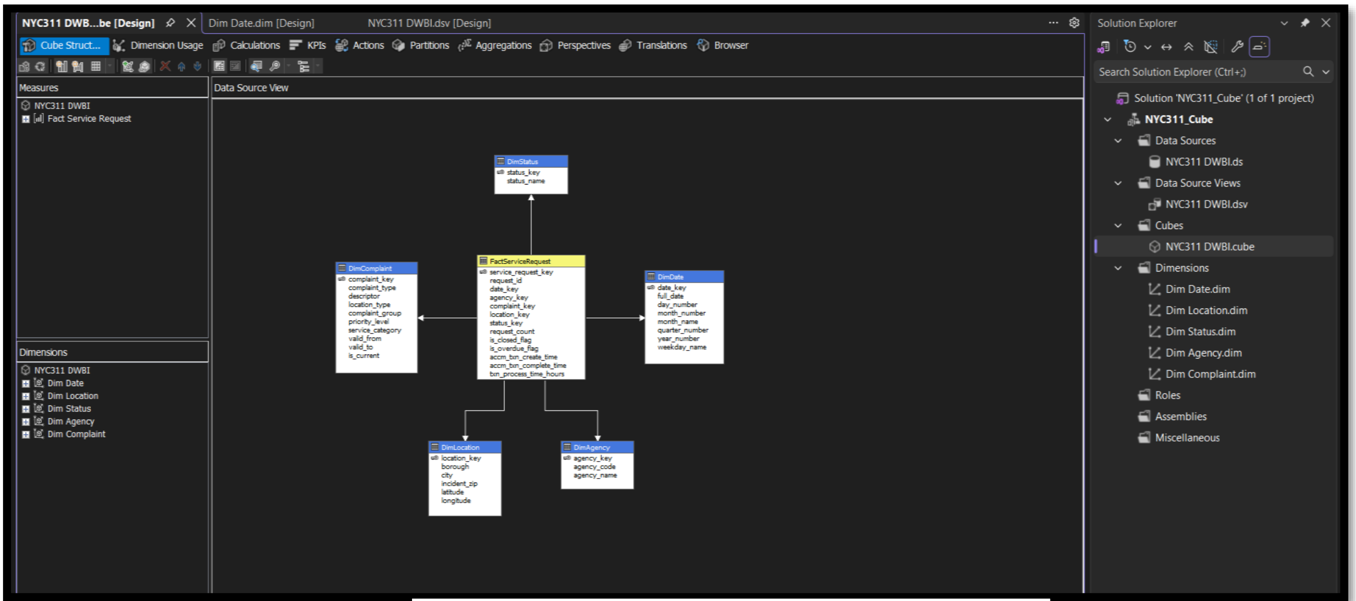
- Fact Table connected to multiple dimensions
- Measures defined for analysis
- Time hierarchy implemented
- Cube successfully deployed and processed



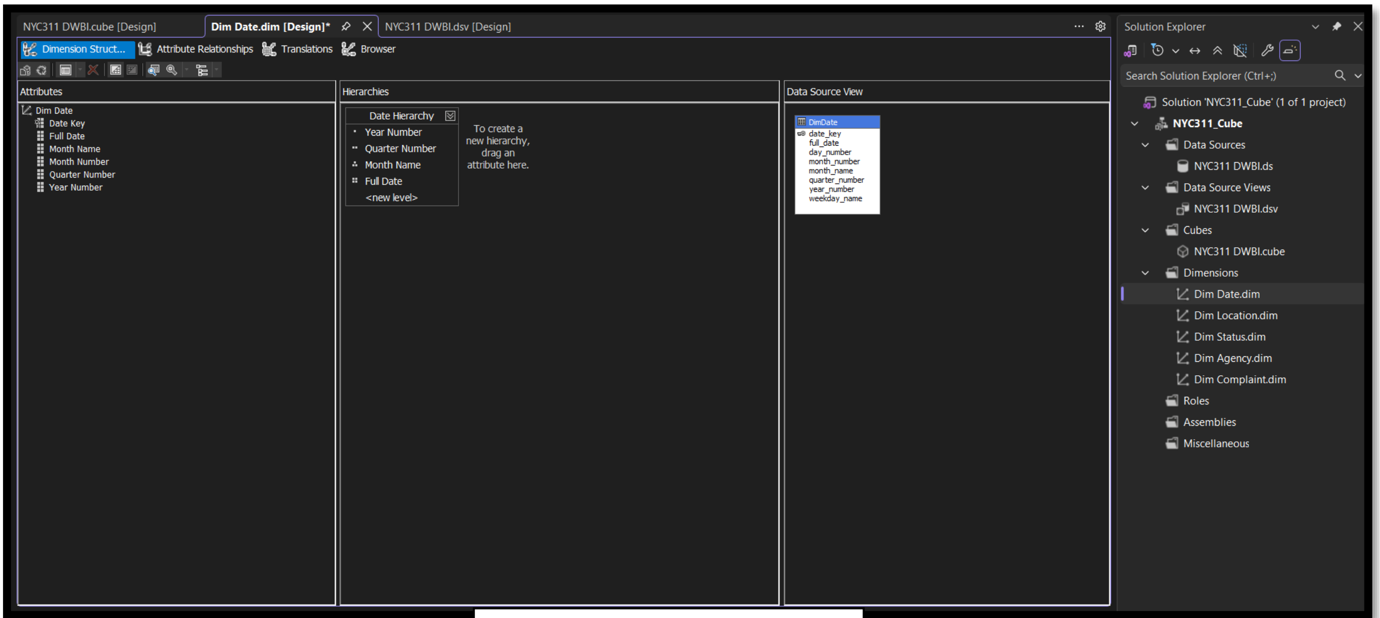
Data Source Configuration



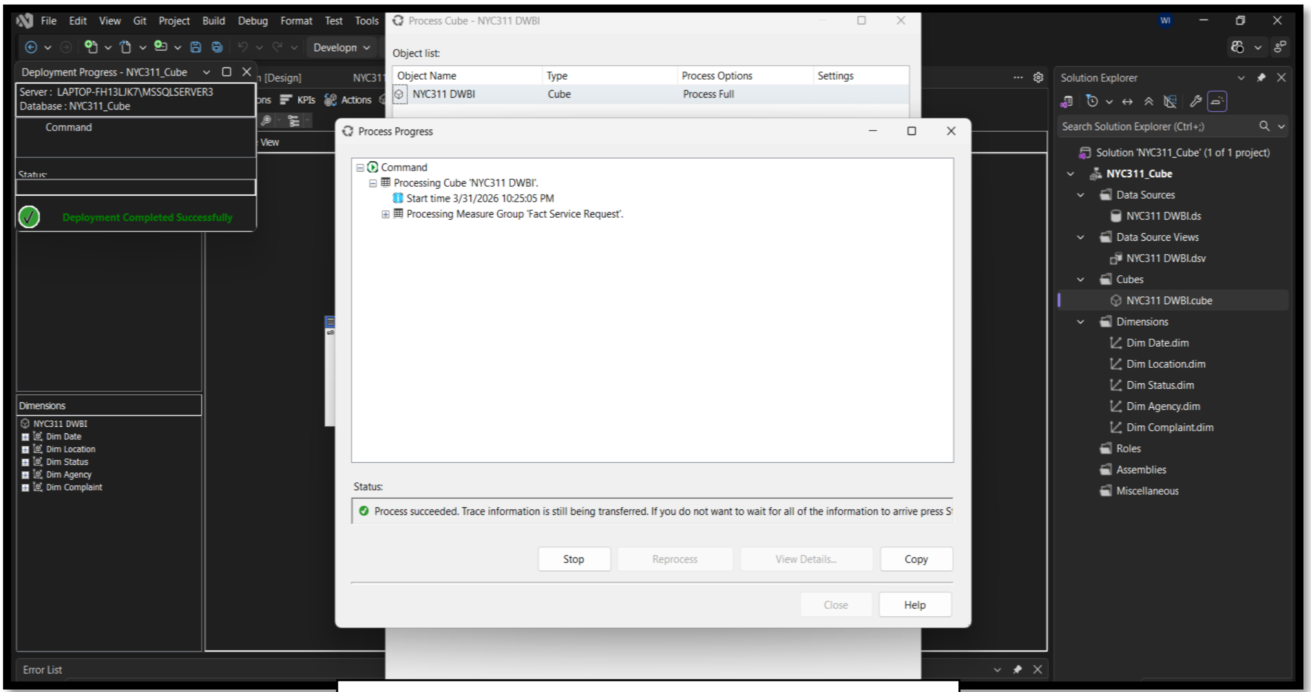
Data Source View showing Fact and Dimension Tables



Cube Structure with Measures and Dimensions



Date Hierarchy in Dim Date



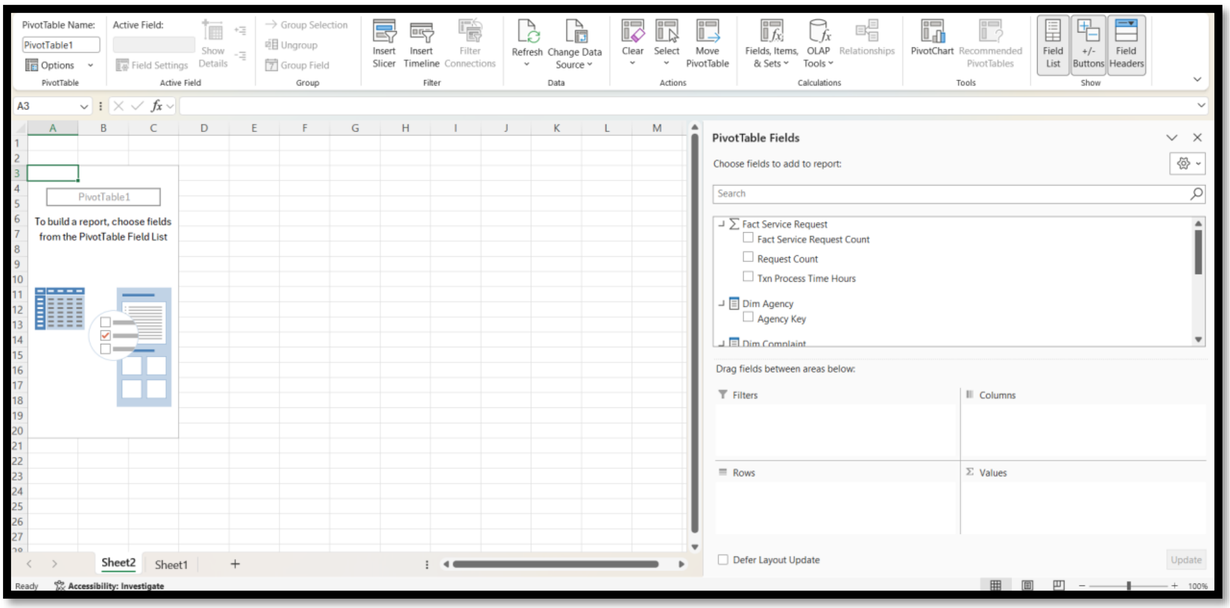
Cube Deployment and Processing Successful

Step 3: Demonstration of OLAP Operations

3.1: Excel connection to the SSAS cube using Analysis Services

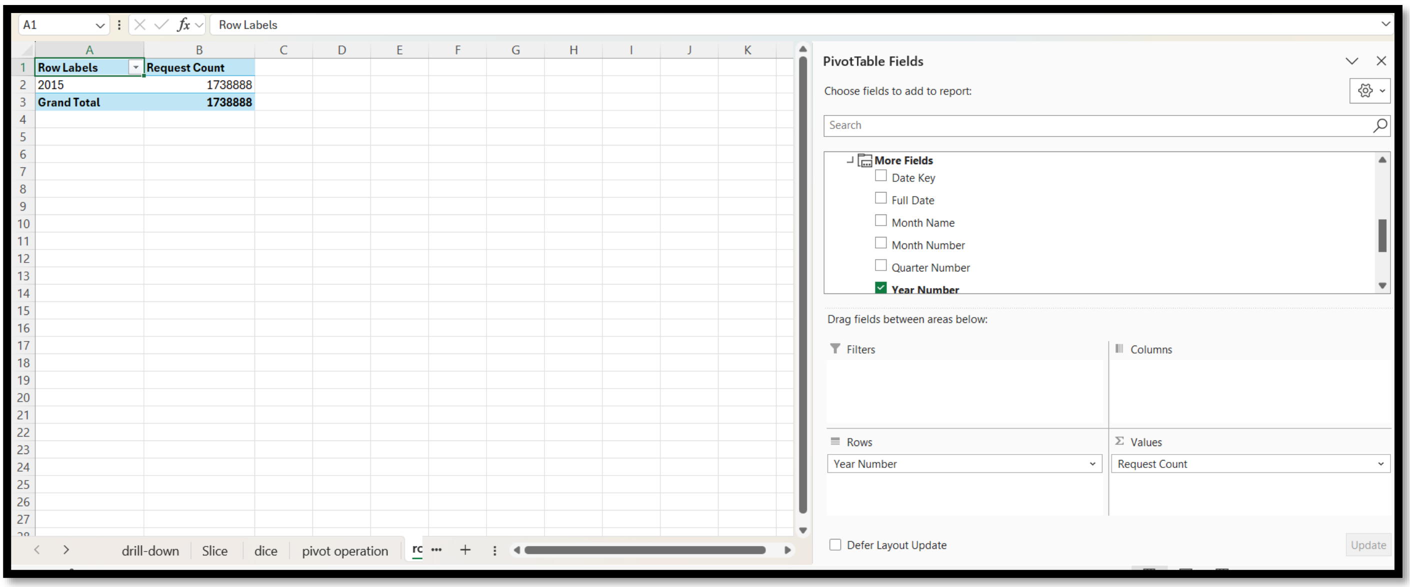
The Excel workbook was connected to the SSAS cube using the "From Analysis Services" option in the Data tab. The server name and cube were selected, and a PivotTable was created to enable data analysis.

This connection allows Excel to retrieve multidimensional data from the cube, enabling OLAP-based analysis using PivotTables and charts.



3.2: Roll-up operation showing total request count at the Year level

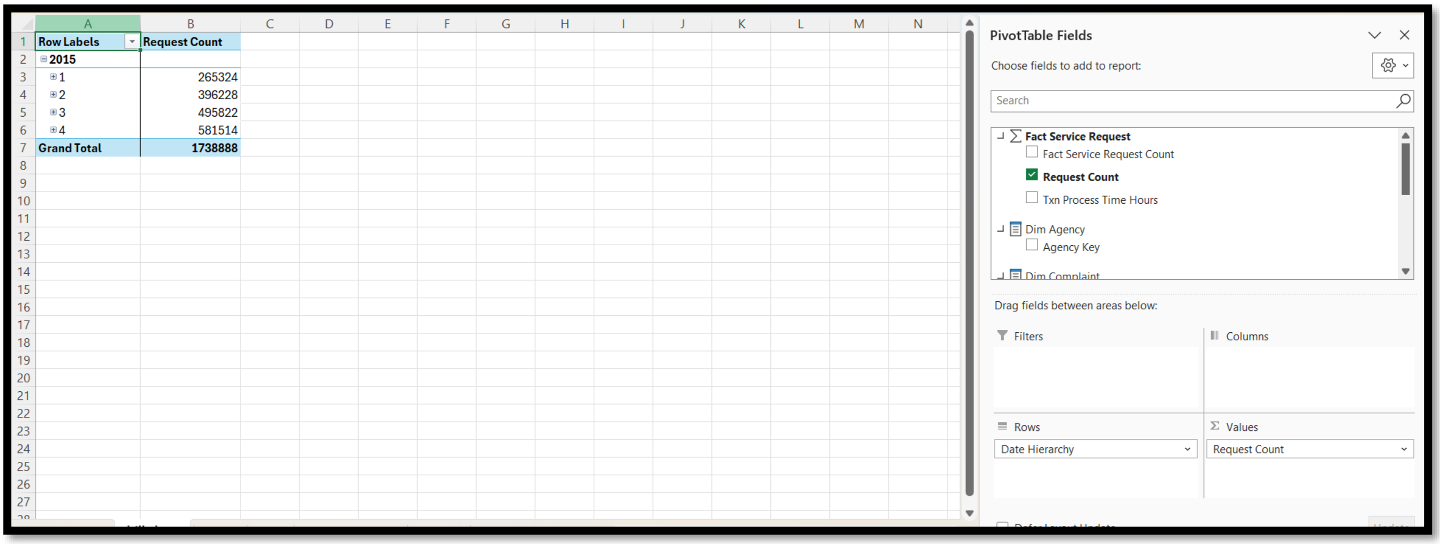
A roll-up operation was performed by placing the Year Number attribute from the Date dimension into the Rows area and Request Count into Values. This aggregates detailed data into a higher level (year level), showing total request counts per year. This helps users understand overall trends and total volume of service requests over time.



3.3: Drill-down operation from Year to Quarter in the Date Hierarchy

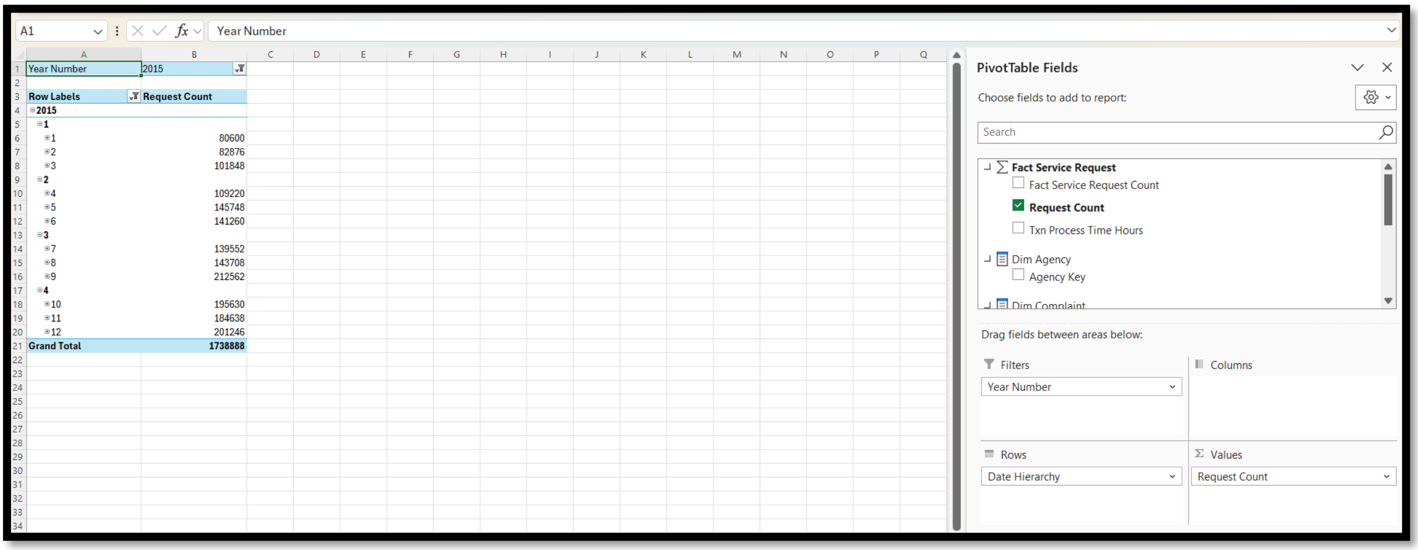
The drill-down operation was performed using the Date Hierarchy (Year -> Quarter -> Month). By expanding the hierarchy, detailed data at lower levels were revealed.

This operation allows users to move from summarized data to more detailed levels. It helps in identifying patterns such as which months or quarters have higher request volumes.



3.4: Slice operation by filtering data for Year 2015

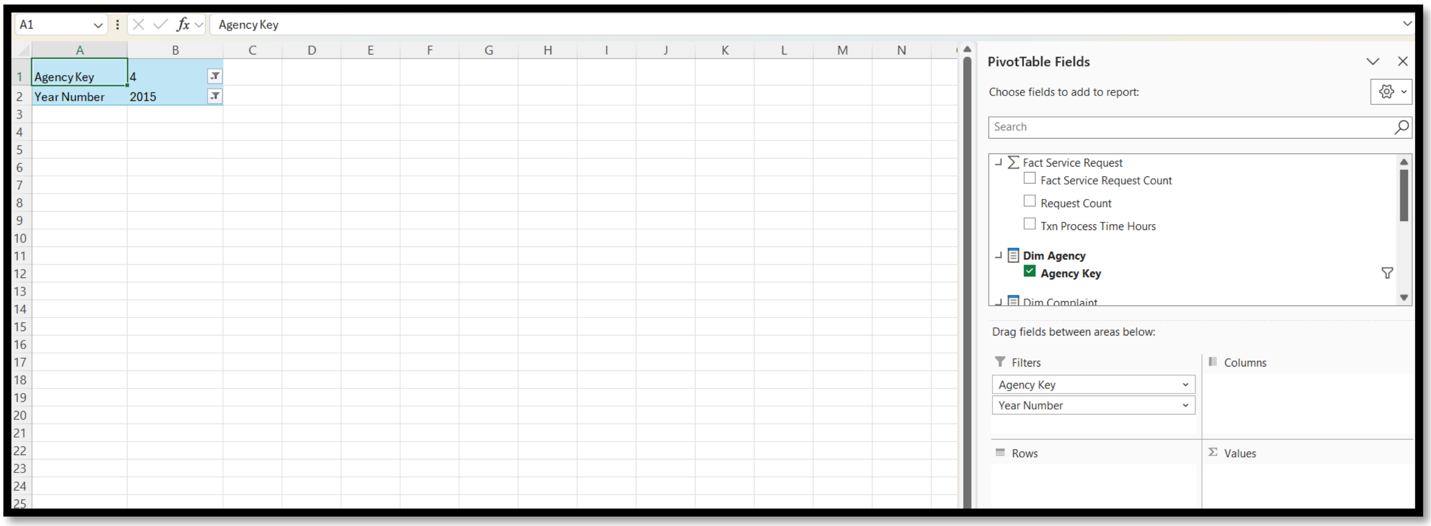
The slice operation was performed by applying a filter on the Year Number (2015). This isolates a single dimension value, allowing analysis of data for a specific year only. It helps users focus on a particular time period without interference from other data



3.5: Dice operation using multiple filters such as Year and Agency

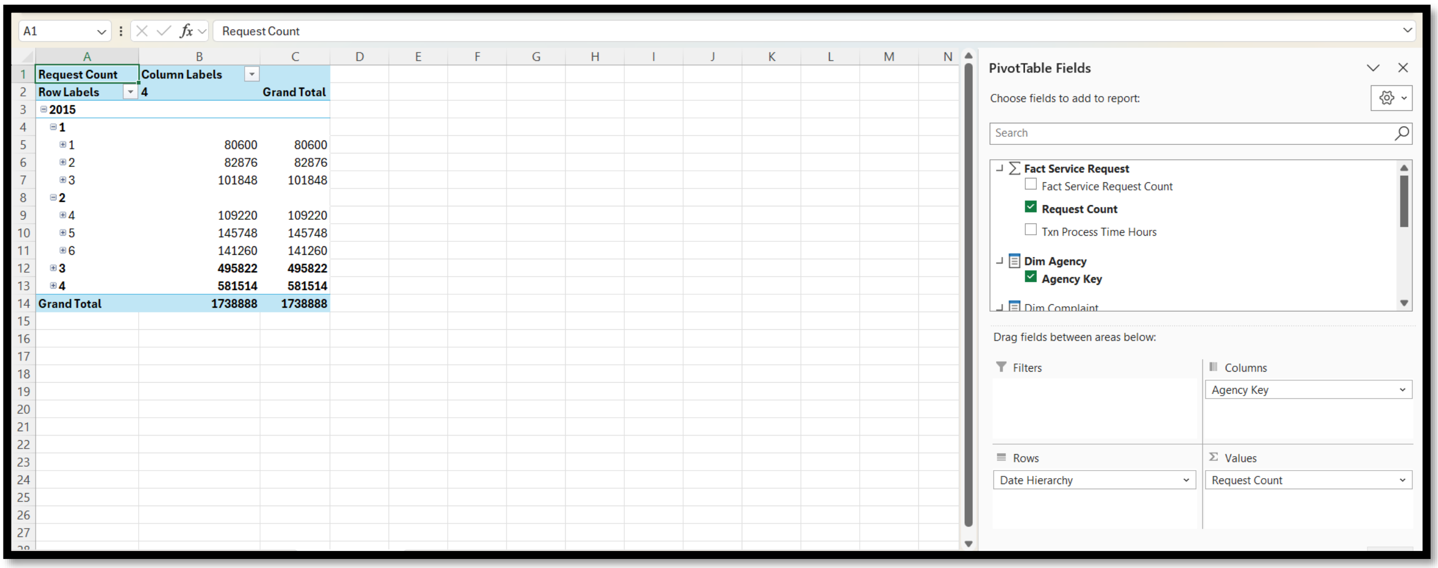
The dice operation was performed by applying multiple filters, including Year Number (2015) and Agency Key (4). This filters the dataset based on multiple dimensions simultaneously.

It allows users to analyze a specific subset of data, providing more focused and meaningful insights.



3.6: Pivot operation showing Request Count by Date Hierarchy and Agency Key

The pivot operation was performed by placing the Date Hierarchy in the Rows area and Agency Key in the Columns area. This rearranges the data orientation, allowing comparison of request counts across different dimensions. It helps users analyze relationships between time and agency performance from different perspectives.



Step 3 was completed by connecting Microsoft Excel to the SSAS cube and creating PivotTables to demonstrate standard OLAP operations. Roll-up was used to view summarized yearly data, drill-down was used to expand the time hierarchy into quarters, slice was used to filter by a single year, dice was used to filter by both year and agency, and pivot was used to rearrange the data layout for comparison. These operations proved that the cube supports multidimensional analysis.

Step 4: PowerBI Reports

Report 1: Creating a report with a Matrix visual to display detailed tabular data with row and column groupings.

1. Data Preparation

The dataset was imported into Power BI from the SQL Server database containing service request records. The data consisted of a central fact table (FactServiceRequest) and multiple dimension tables including DimDate, DimAgency, DimComplaint, DimLocation, and DimStatus.

The dataset was cleaned and validated to ensure:

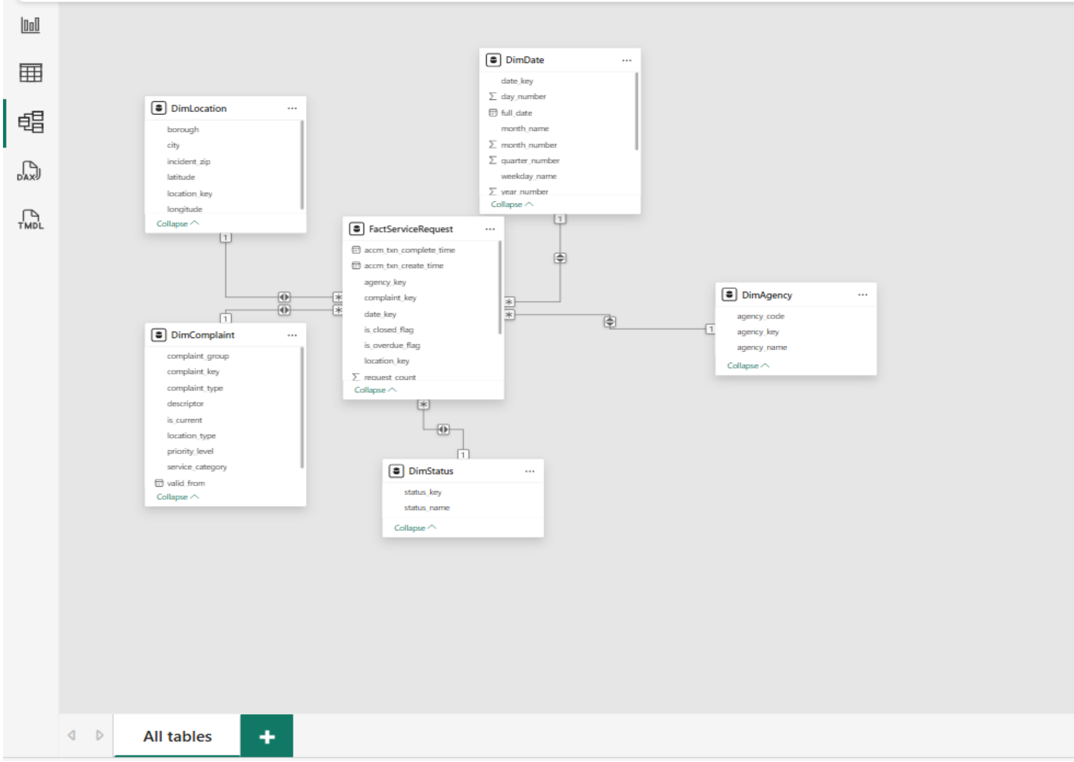
- No missing key values
- Proper data types (numeric, text, date)
- Consistent relationships between tables

2. Data Modeling

A star schema was implemented:

- FactServiceRequest (fact table)
- Connected to:
    - DimDate via date_key
    - DimAgency via agency_key
    - DimComplaint via complaint_key
    - DimLocation via location_key
    - DimStatus via status_key

Relationships were defined as one-to-many (1: *), ensuring proper filtering and aggregation across dimensions.



3. Visual Design

A Matrix visual was created to display multidimensional data:

- Rows: year_number (from DimDate)
- Columns: agency_name (from DimAgency)
- Values: Total Requests

This allows users to analyze request volumes across agencies and time.

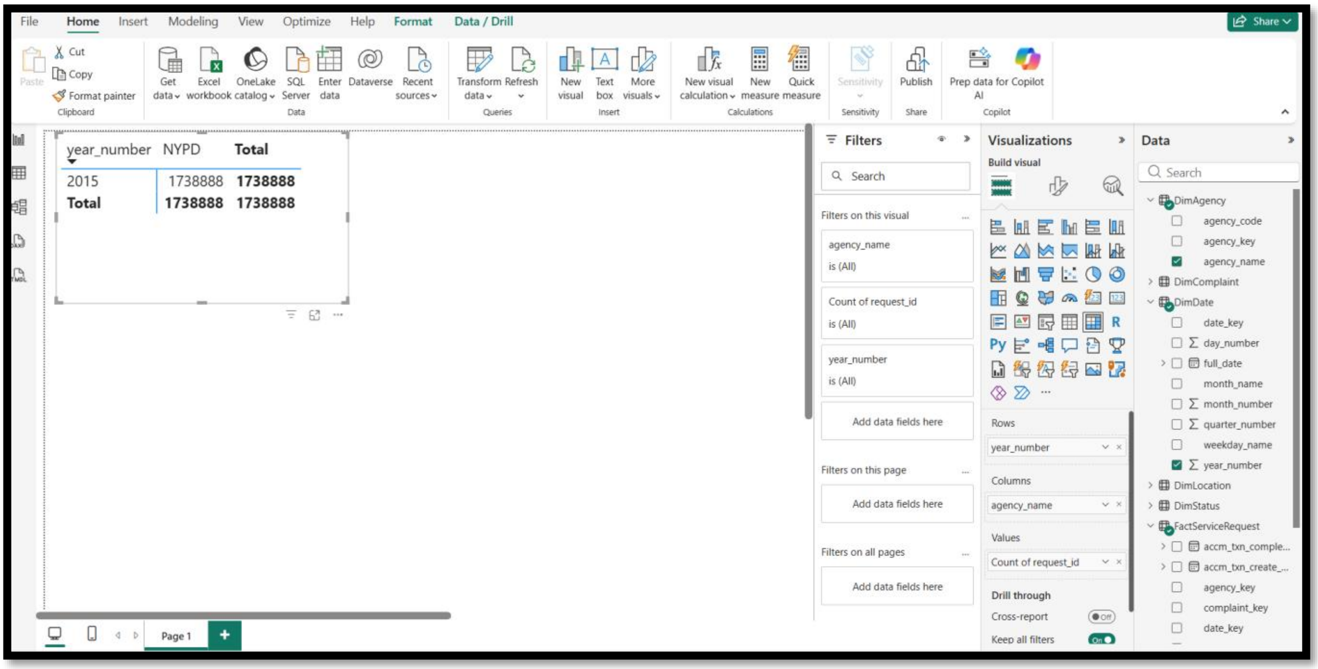
The matrix provides:

- Structured tabular view
- Clear grouping of rows and columns
- Easy comparison between categories

4. Insights

The matrix visual enables users to:

- Compare request volumes across agencies
- Analyze yearly trends
- Understand data distribution in a structured format



Report 2: Creating a report with multiple slicers

1. Data Preparation

The same dataset and model from Report 1 were reused to maintain consistency. All dimension tables were properly connected to the fact table to allow filtering.

2. Slicers (Parameters)

Two primary slicers were created using attributes from different dimension tables:

- Status Name from the DimStatus table
- Complaint Type from the DimComplaint table

These slicers were connected through the relationships defined in the data model, allowing them to interact with each other.

When a user selects a value in the Status Name slicer (e.g., Open, Closed, Assigned), the available values in the Complaint Type slicer are dynamically filtered to display only relevant complaint categories associated with the selected status. This demonstrates the implementation of cascading filtering.

3. Visualizations Used

Multiple visuals were created to represent insights from the dataset

1. Bar Chart

- Displays the count of service requests by complaint type
- Helps identify the most frequent types of complaints

2. Pie Chart

- Shows the distribution of service requests by status
- Provides a clear overview of request completion and progress

3. Line Chart

- Displays the trend of service requests over time (monthly)
- Helps analyze patterns and fluctuations across different months

All visuals use the count of request_count from the fact table as the measure.

4. Interactivity and Filtering

The report supports full interactivity:

- Selecting a value in any slicer filters all visuals simultaneously
- The slicers are interconnected through data model relationships
- Cascading behavior ensures dependent filtering between slicers

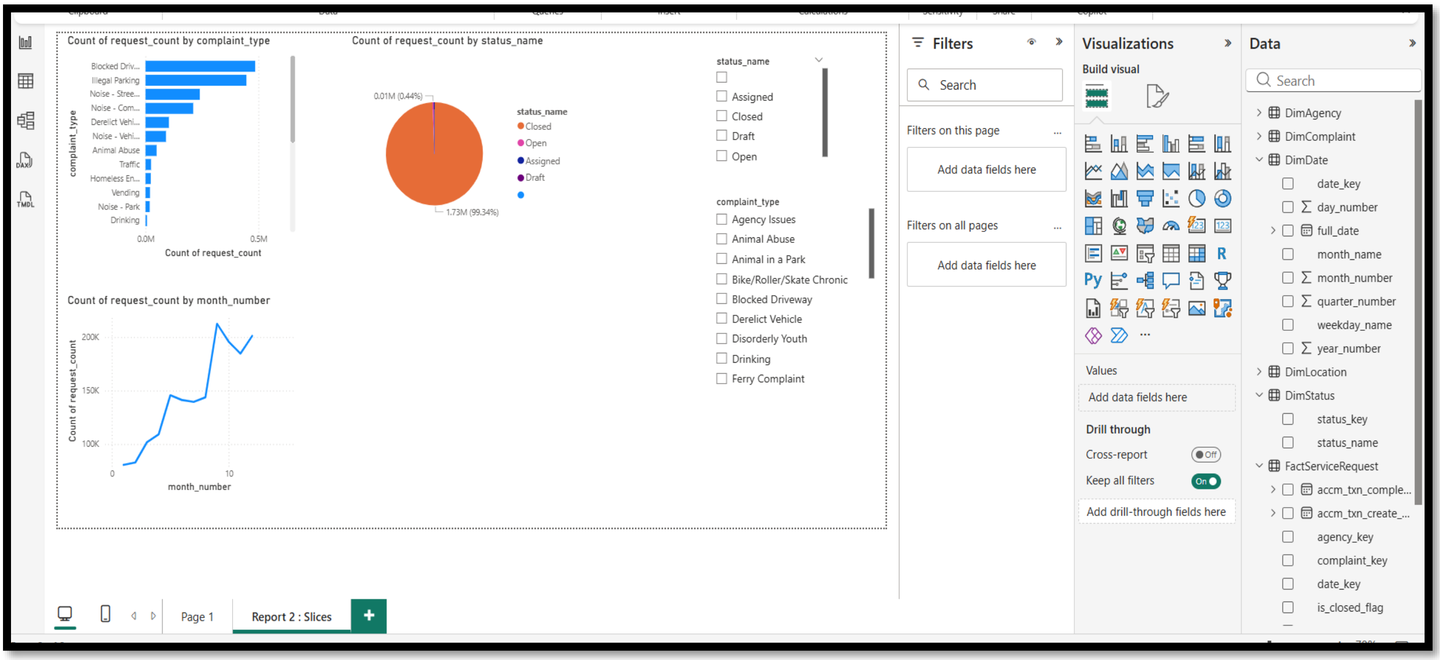
5. Data Model Integration

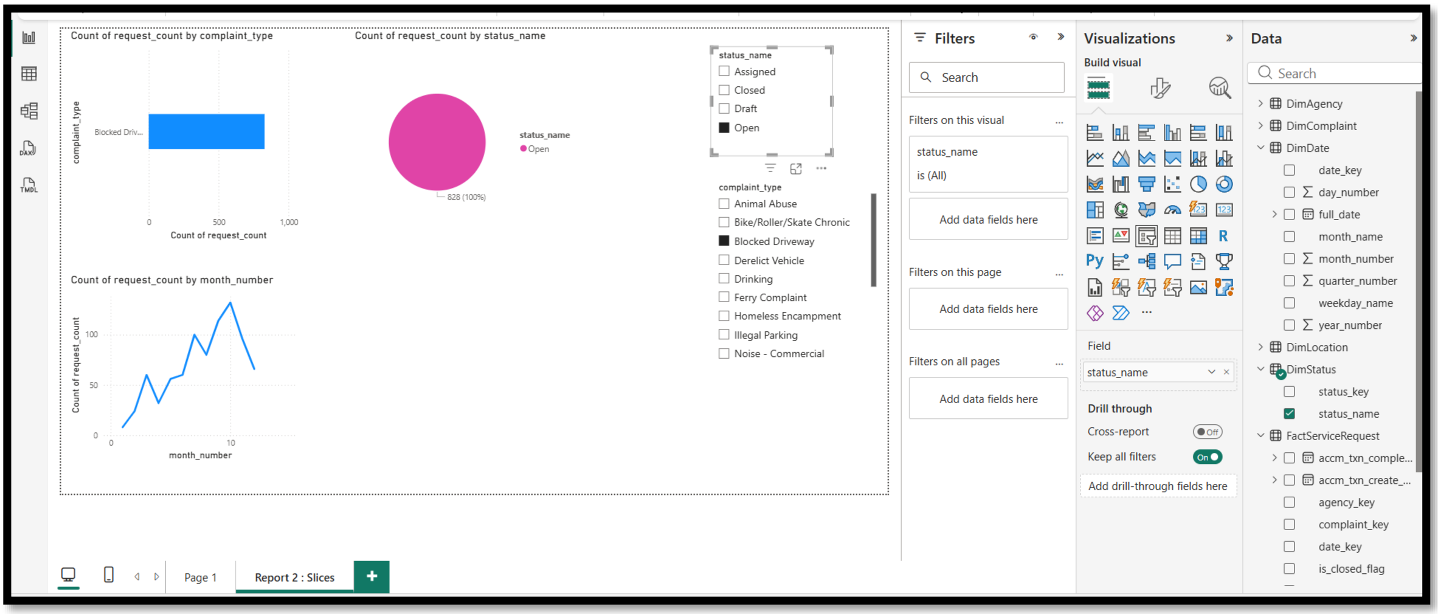
The cascading filtering functionality is enabled through relationships between:

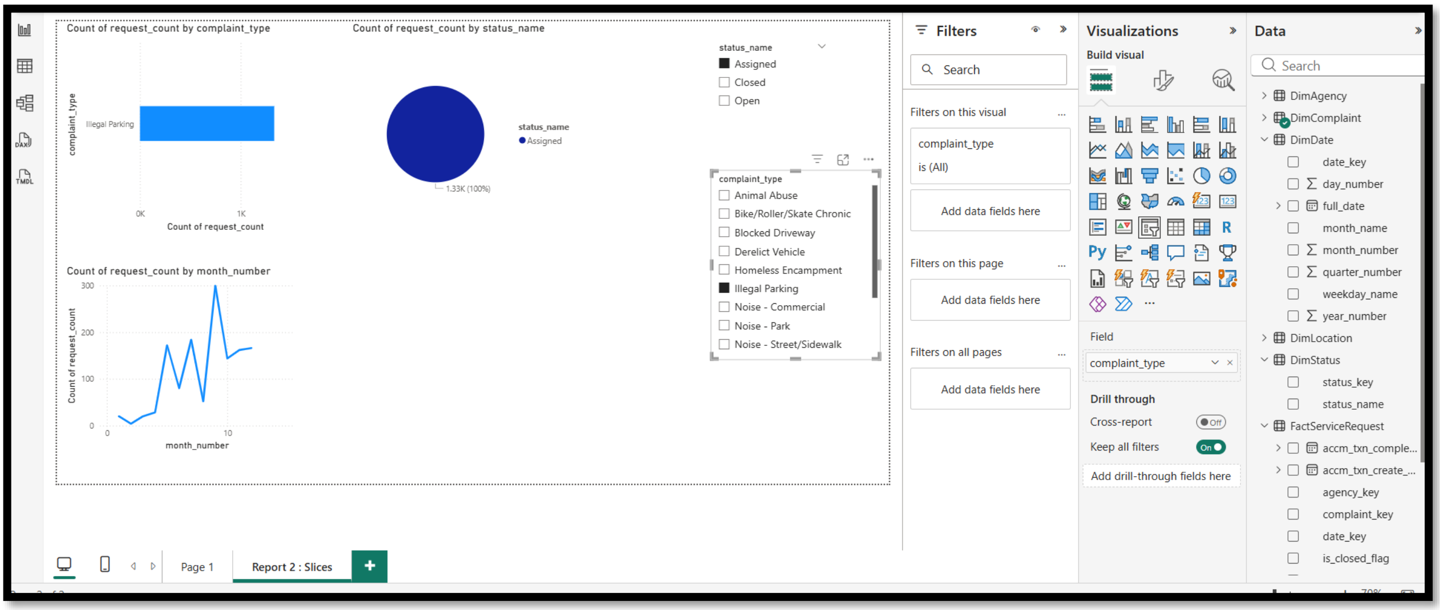
- FactServiceRequest (fact table)
- DimStatus (status dimension)
- DimComplaint (complaint dimension)
- DimDate (time dimension)

These relationships allow filters applied on dimension tables to propagate to the fact table and across other dimensions.

This demonstrates how Power BI can be used to create interactive dashboards with cascading slicers. The combination of multiple visuals and dynamic filtering enhances data exploration and provides meaningful insights into service request patterns


Count of request_count by complaint_type
Count of request_count by status_name
Count of request_count by month_number
status_name
Assigned
Closed
Draft
Open
complaint_type
Agency Issues
Animal Abuse
Animal in a Park
Bike/Roller/Skate Chronic
Blocked Driveway
Derelict Vehicle
Disorderly Youth
Drinking
Ferry Complaint
Filters
Visualizations
Data
Page 1
Report 2 : Slices


Count of request_count by complaint_type
Count of request_count by status_name
Count of request_count by month_number
status_name
Assigned
Closed
Draft
Open
complaint_type
Animal Abuse
Bike/Roller/Skate Chronic
Blocked Driveway
Derelict Vehicle
Drinking
Ferry Complaint
Homeless Encampment
Illegal Parking
Noise - Commercial
Filters
Visualizations
Data
Page 1
Report 2 : Slices


Count of request_count by complaint_type
Count of request_count by status_name
Count of request_count by month_number
status_name
Assigned
Closed
Open
complaint_type
Animal Abuse
Bike/Roller/Skate Chronic
Blocked Driveway
Derelict Vehicle
Homeless Encampment
Illegal Parking
Noise - Commercial
Noise - Park
Noise - Street/Sidewalk
Filters
Visualizations
Data
Page 1
Report 2 : Slices

Report 3: Creating a drill-down report that allows users to explore data hierarchically

1. Data Preparation

The dataset was reused with focus on the Date dimension hierarchy

2. Data Modeling

The DimDate table provides hierarchical fields:

- Year
- Quarter
- Month

These are connected to the fact table via date_key

3. Visual Design

A Clustered Column Chart was created with:

- X-axis: Date Hierarchy (Year → Quarter → Month)
- Y-axis: Total Requests

Drill-Down Features

Power BI's built-in drill-down functionality was used:

- Year level -> overview
- Quarter level -> breakdown
- Month level -> detailed view

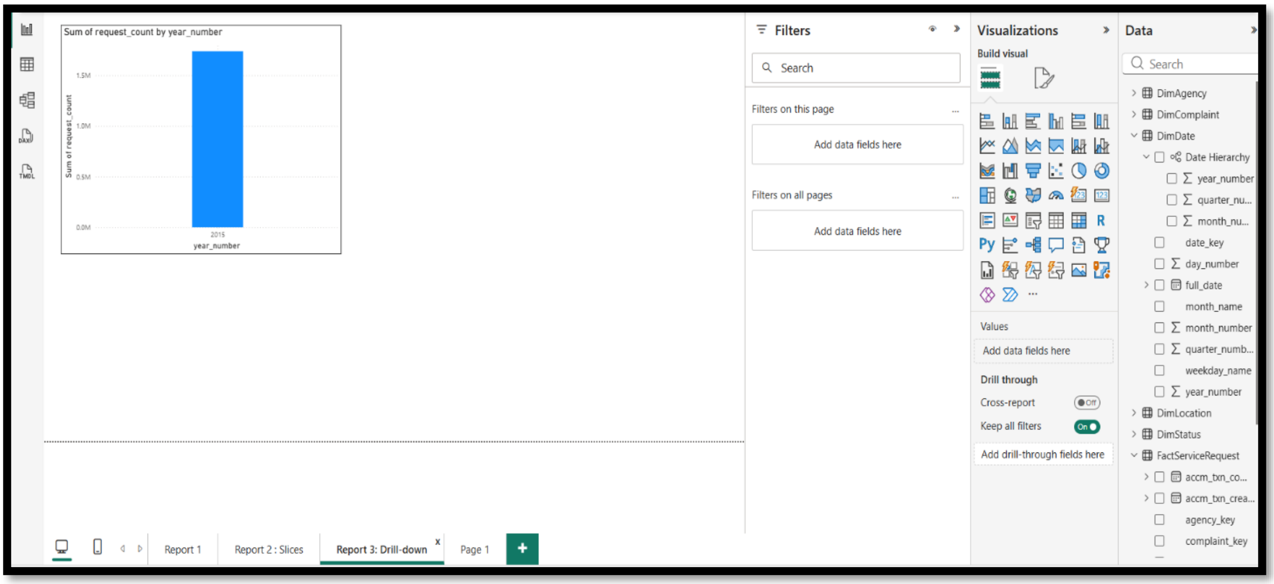
Users can:

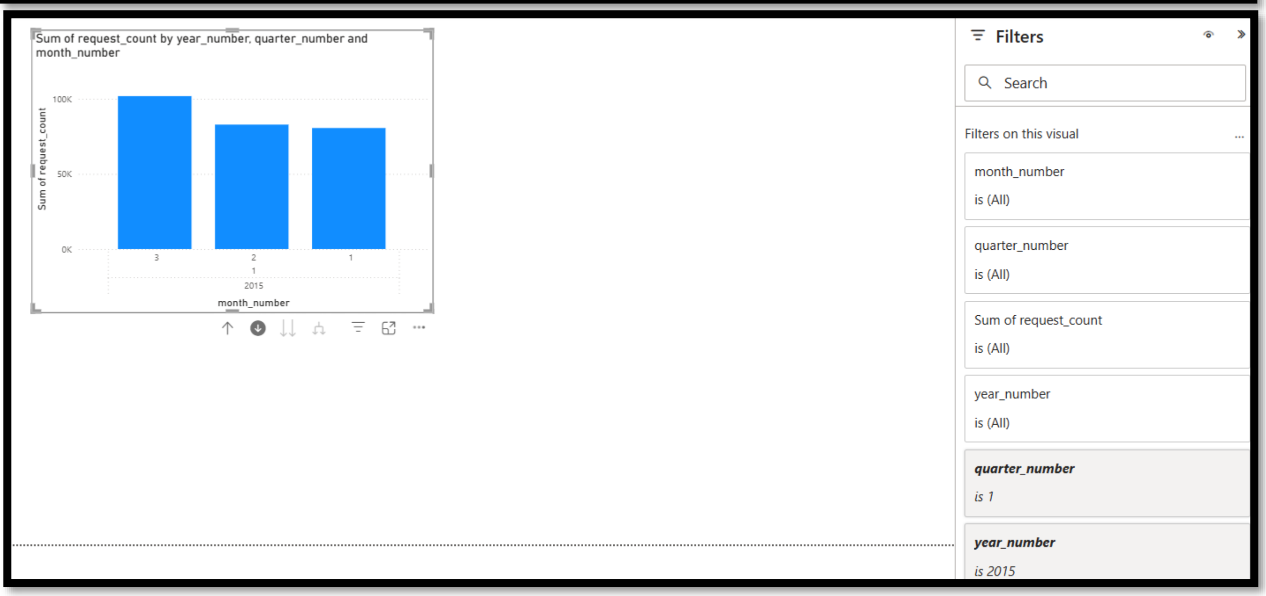
- Click drill-down buttons
- Navigate between hierarchy levels
- Analyze trends progressively

5. Data Limitation

The dataset contains data for only one year (2015). Therefore:

- No year-to-year comparison is possible
- Drill-down focuses on intra-year analysis





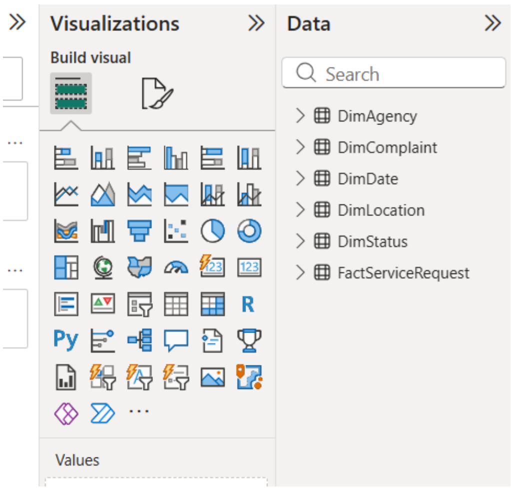
Report 4: Create a drill-through report

1. Data Preparation

The dataset created in Assignment 1 was reused in Power BI. The dataset includes:

- Fact table: FactServiceRequest
- Dimension tables:
    - DimAgency
    - DimComplaint
    - DimStatus
    - DimDate
    - DimLocation

The dataset was loaded from SQL Server and prepared for analysis.



2. Data Modeling

A star schema model was used where the fact table is connected to dimension tables.

The drill-through functionality mainly uses:

- DimAgency (agency_name)

This is connected to the fact table using:

- agency_key

This relationship allows filtering of detailed data based on the selected agency.

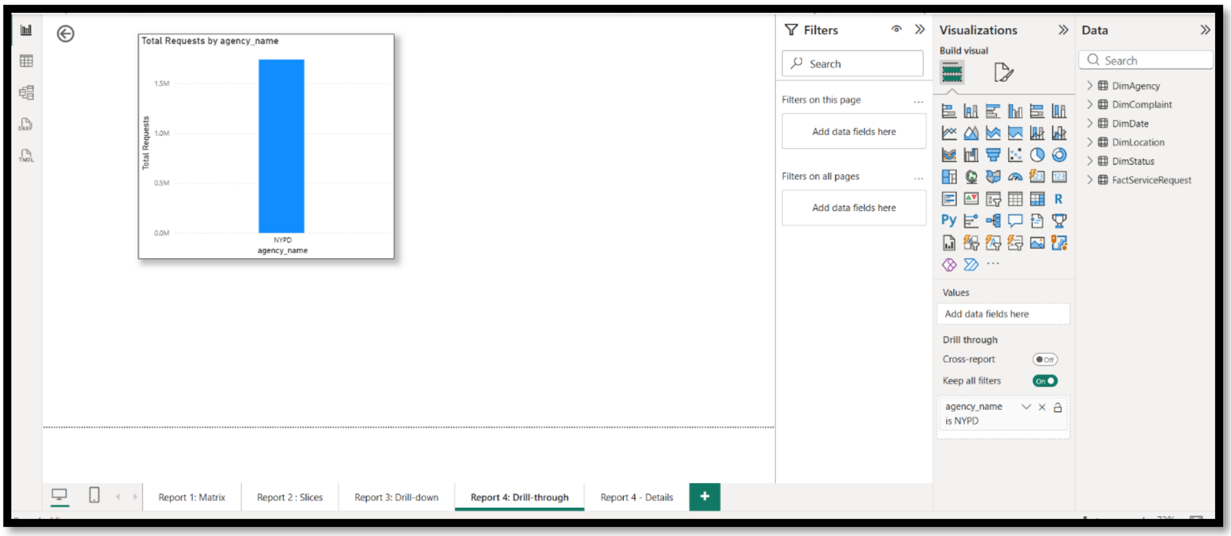
4. Visual Design

Main Page (Summary View)

A clustered column chart was created with:

- X-axis: agency_name
- Y-axis: Total Requests

This chart provides a summary of request counts by agency.



5. Drill-Through Page (Detailed View)

A separate page named "Report 4 – Details" was created.

The field: agency_name

was added to the Drill-through section to enable filtering.

6. Detail Visuals

The detail page includes Table Visual

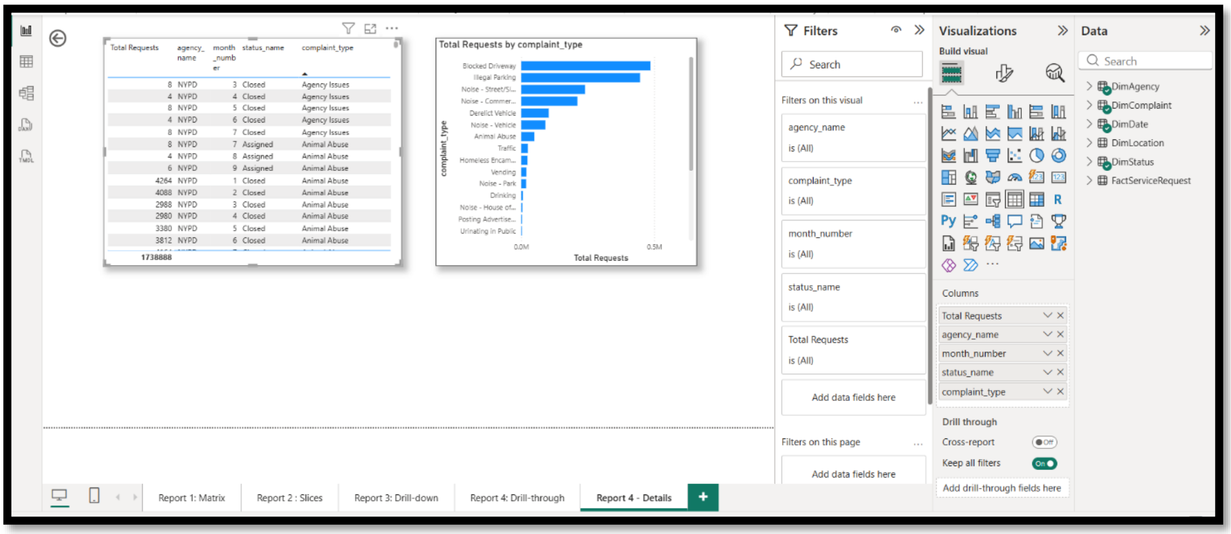
Displays:

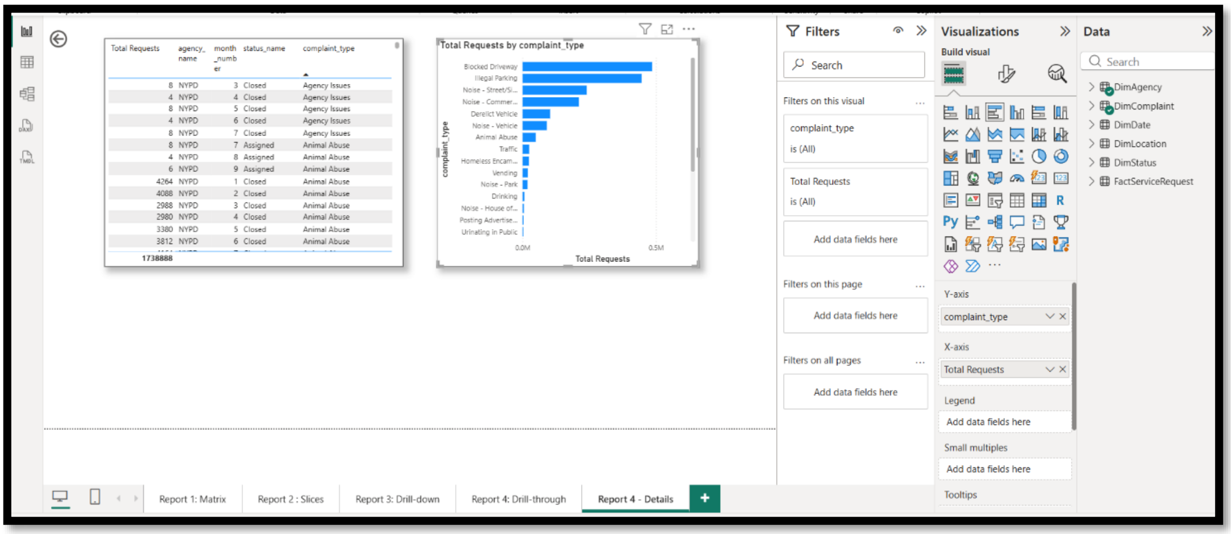
- agency_name
- complaint_type
- status_name
- month_number
- Total Requests

Bar Chart

- Axis: complaint_type
- Values: Total Requests

These visuals show detailed information related to the selected agency.





7. Insights

The drill-through report allows users to:

- Move from summary-level data to detailed information
- Analyze specific categories (e.g., agency) in depth
- Understand relationships between complaint type, status, and time

This improves usability and supports better analytical decision-making